

Pressure dynamics in the bottleneck flow of self-propelled particles

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(Received 24 June 2025; accepted 15 December 2025; published xxxxxxxxxx)

We present an experimental investigation of the pressure dynamics during the flow of self-propelled particles through narrow passages. When the ensemble is flowing, pressure fluctuates around a constant value that does not depend on the crowd size, suggesting that the orifice locally determines the dynamics in this scenario. On the contrary, when the system clogs, pressures are higher for larger crowd sizes, highlighting the importance of the whole collectivity in the process. Then, by correlating the pressure evolution with the exit time of the self-propelled particles, we discover that when a clog is resolved, pressure suddenly drops as a consequence of system reorganization. After this dramatic event, there is a sustained pressure growth over time that shows a square-root dependence, compatible with the structural aging that has been proposed to be behind the broad tail distributions of clogging times.

DOI: [10.1103/gqmg-t16q](https://doi.org/10.1103/gqmg-t16q)

I. INTRODUCTION

When a system composed by discrete bodies passes through a bottleneck the flow may become interrupted due to the formation of clogs that arrest the flow [1]. Then, the clog can be resolved, meaning that the blockage at the outlet is broken and particle flow resumes, either through the application of an external perturbation (if the particles are inert) [2,3] or through the agents' intrinsic activity (if they are active) [4,5]. The successive alternation of these two processes—clogging and unclogging—leads to the characteristic intermittent flow observed in these scenarios, which range from the apparently simple silo discharge to pedestrian evacuation through a narrow gate [6]. In general, the flow in all these systems alternates between periods of arrest and bursts of continuous passage of particles. The durations of the bursts (bursting times) typically follow an exponential distribution, whereas the clog durations display a broad tail, sometimes compatible with a power law, $\text{PDF}(\tau) \propto \tau^{-\gamma}$. Indeed, when $\gamma \leq 2$, the first moment of the distribution (the average of clogging time) does not converge, a feature that has been used to define the clogging transition in a rigorous way [7]: the system is clogged for $\gamma \leq 2$ and unclogged (but generally with intermittent flow) for $\gamma > 2$.

The exponential distribution of the bursting times can be explained if the clog formation is a memoryless process with constant occurrence probability throughout all the duration of the burst [8–12]. However, the origin of the broad tail distribution of clogging times is not as well understood and two different theories have been proposed in the last years. The first one relied on the different energies necessary to resolve each of the configurations that are able to block a given geometrical configuration [13]. The second attributed the broad

tails to an aging process of the clogging configuration that would lead to a reduction of the unclogging probability with time [14,15].

Interestingly, the only exceptions for the broad tail distribution in the unclogging times have been found in the flow of fish through an orifice [16] and pedestrian evacuations while keeping social distancing [17]. In both scenarios, contacts among particles do not exist, and therefore there is no buildup of pressure within the system. It is precisely the pressure—an ingredient that seems fundamental to understanding the clog destruction process—one of the variables whose behavior has been least studied in bottleneck flows.

So far, we know that clogs last longer when increasing either the driving force [18,19] or the crowd size [5]; both parameters are, in principle, associated with the amount of pressure built up within the system. A similar behavior has been observed in suspensions driven at constant pressure, where clogs persist longer under higher-pressure gradients. However, when the flow rate is imposed instead, the probability of breaking a clogging arch becomes practically independent of its magnitude [19,20]. Beyond some preliminary results for highly competitive pedestrian flows [21] and measurements in silos under continuous discharge conditions [22], little is known about the dynamics of pressure during intermittent flow.

In this work, we address this gap by experimentally investigating the pressure dynamics in the intermittent flow of self-propelled particles through a bottleneck. The key difference of this study is the temporal measurement of mechanical pressure during clogging events of active particles. Our system consists of commercial vibration-driven vehicles (Hexbug Nano) acting as active agents in a quasi-two-dimensional arena where flow intermittency has been previously reported [5,23]. We show that pressure fluctuations reveal clear signatures of the system's phase state, allowing us to distinguish

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between unclogged and clogged states. Moreover, the slow, square-root buildup of pressure after the release of an agent is consistent with a structural aging process proposed to explain the broad, power-law distribution of clogging durations.

The experimental setup and measurement procedures are described in Sec. II, the results and their analysis are presented in Sec. III, and the main conclusions and implications are discussed in Sec. IV.

II. EXPERIMENTAL SETUP

We used commercially available vibration-driven vehicles (VDVs), known as Hexbug Nano, as an example of an inertial self-propelled agent [24–27]. Each vehicle measures $43 \times 15 \times 18$ mm, and its motion is generated by the vibration of an eccentric motor, rectified by a brushlike structure that acts as the legs. The motor is powered by a 1.5-V battery. The direction of motion of these vehicles is influenced by collisions with other vehicles and can also be guided by the geometry of the container. The VDV's were placed inside an arena shaped similarly to a silo [see Fig. 1(a) for a snapshot including the characteristic dimensions]. The arena had two openings: one exit located at the end of a funnel-like section, and another on the opposite side through which the vehicles were manually reintroduced after exiting. The arena was made of wooden walls lined with a polyvinyl chloride (PVC) sheet and magnetically attached to a metallic base plate. A glass sheet was placed 19 mm above the surface to prevent the vehicles from overturning. The exit opening was $L = 30$ mm wide, corresponding to the width of two VDV's. To measure the pressure exerted by the VDV's on the exit opening, we used two independent wooden sections [highlighted in yellow in Fig. 1(a)] that were connected to the main frame through two load cells (highlighted in blue). The end sections were 15 mm thick, ensuring that they did not touch either the base or the glass cover, so the forces exerted by the VDV's on them were transmitted to the load cells. The load cells had a measurement range of ± 1 kg. Both were connected to a 24-bit analog-to-digital converter, which allowed sampling at a rate of 44 S/s. To track the trajectories of the vehicles and determine the time at which each VDV exited the arena, we used a top-view digital video camera operating at 24 frames/s (FPS). Vehicle identification was achieved using ArUco markers [28,29]: each VDV was equipped with a unique tag, allowing for unambiguous particle tracking [30].

Each experiment lasted more than 1 h and was stopped once the VDV's exiting the arena began to show signs of battery depletion. In each run, we varied the number of VDV's in the arena: $N_{\text{tot}} = 30, 40, 45$, and 50. Throughout the experiment, both video recordings and force measurements were continuously acquired to track the system's evolution and monitor the interaction forces at the outlet. To avoid battery drain and ensure the continuity of the experiments, clogging events lasting more than 2 min were manually resolved by removing one of the VDV's involved in the clog. Importantly, those clogs were annotated and added to the survival distribution that will be presented below as clogs longer than 120 s (indeed, this is the reason for which this distribution cannot be represented beyond this value). The analysis presented in

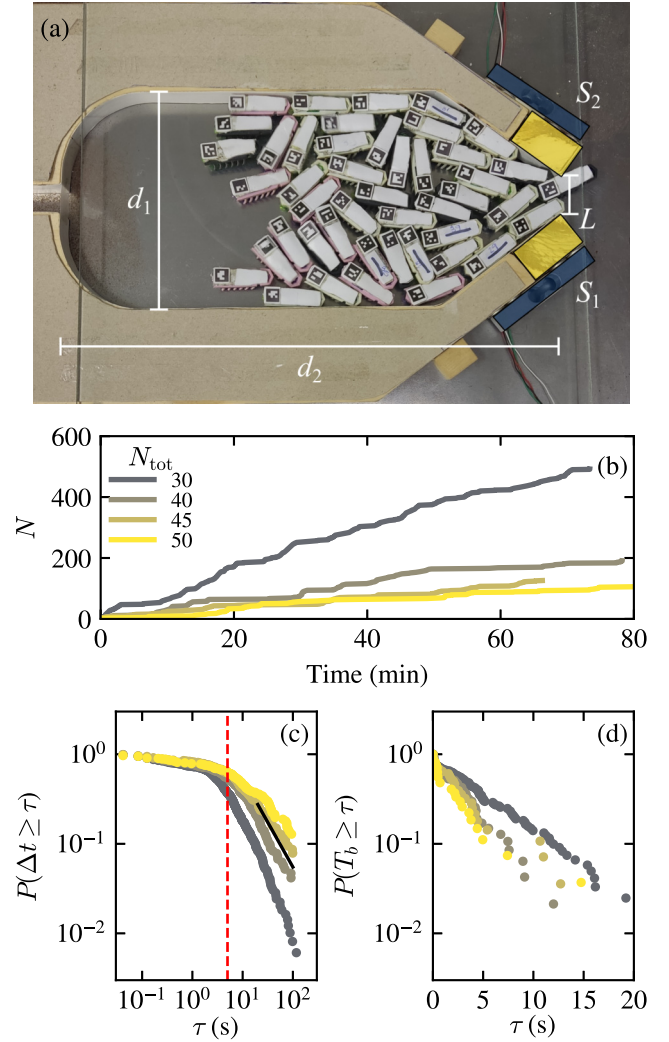


FIG. 1. (a) Experimental setup. Particles are introduced in the arena from the left, accumulate and exit through the small aperture of size $L = 30$ mm at the right. This exit is made by two wooden sections (marked in yellow), on which the forces are measured by two force sensors, S_1 and S_2 , marked in blue. The characteristic dimensions of the system are also displayed: $d_1 = 155$ mm, and $d_2 = 350$ mm. Each VDV is identified with a unique ArUco marker placed on its back. (b) Number of evacuated agents as a function of time for different numbers of total VDV within the arena as indicated in the legend. (c) Survival function of Δt , the time interval between consecutive VDV exits. Note the logarithmic scale in both axis. The vertical line marks the threshold separating the bursting and clogging regimes. The solid black line indicates a reference trend corresponding to $P(\Delta t \geq \tau) \propto \tau^{-1}$. (d) Survival function of the burst duration for the four crowd sizes. The color code is the same as in (b). Note the logarithmic scale only in the y axis.

this work was performed using the data collected before each experiment was stopped due to battery depletion.

III. RESULTS

As a first step in our analysis, we examined how the total number of agents influences the outflow dynamics. To this end, we tracked the cumulative number of agents, N , exiting

the arena over time for each experimental configuration, as shown in Fig. 1(b). As previously reported in Ref. [5], increasing the number of agents in the arena leads to a reduced outflow rate. This behavior has been attributed to longer-lasting clogs, likely caused by the increased pressure within the system.

Effectively, the analysis of the distributions of headway times, i.e., time intervals between consecutive exits of VDV (Δt), corroborates that clog duration remarkably enlarges with the crowd size. In particular, in Fig. 1(c) we show the survival functions [or complementary cumulative distribution functions (CCDFs)], of Δt , i.e., the probability that an interval Δt exceeds a given duration τ , $P(\Delta t \geq \tau)$. As the number of VDV in the system increases, the distribution tails become heavier, consistent with power-law behavior, $P(\Delta t \geq \tau) \propto \tau^{-\gamma+1}$, which corresponds to a probability density function of the form $\text{PDF}(\Delta t) \propto \Delta t^{-\gamma}$. To estimate the exponent γ , we used the method introduced by Clauset *et al.* [31,32], which yields both the power-law exponent and the minimum time $\tau_{\min}$ from which the fit is valid. In our experiments, $\tau_{\min}$ ranged from 3 to 11 s, while the power-law exponent γ took the values 2.2 ± 0.2 , 2.1 ± 0.2 , 1.7 ± 0.2 , and 1.5 ± 0.2 for $N_{\text{tot}} = 30, 40, 45$, and 50 , respectively. These results are consistent with those reported in Ref. [5], where it was shown that increasing the number of VDV leads to smaller values of γ , reaching values below 2, a critical point at which the average clogging time diverges. Under the specific conditions of our experiment (outlet size, VDV excitation level, etc.), the transition from the unclogged ($\gamma > 2$) to the clogged ($\gamma \leq 2$) state appears to occur within the range $N_{\text{tot}} = 40\text{--}45$, as illustrated by the black line in the figure, which corresponds to a distribution with $\gamma = 2$. Moreover, the value of $\tau_{\min}$ estimated by the fittings can be used as a threshold to separate the bursting from the arrested intervals of the intermittent flow. In this work, although the values of $\tau_{\min}$ estimated from the power-law fits vary slightly across experimental conditions, we adopt $\tau_{\text{th}} = 5$ s as a general threshold to distinguish between the bursting and arrested intervals in all experiments. The selected threshold is indicated by a vertical dashed line in Fig. 1(c). The distribution of burst durations (T_b), defined as the duration of consecutive exit events separated by intervals shorter than τ_{th} , displays an exponential decay [Fig. 1(d)], indicating that the probability of the system becoming clogged remains constant over time. The steepest slopes observed for the cases with more agents in the arena also indicate that the time to reach a clog decreases as N_{tot} increases.

Once we confirmed that our setup reproduces the main features of the intermittent flow previously observed in this system, we turned our attention to the analysis of the pressure measurements. Figures 2(a) and 2(b) show representative temporal evolutions of the force signals recorded by the two sensors, together with the exit times of the VDV (red lines). Two main features emerge from these plots: (i) a clear correlation between the two force signals, and (ii) in many cases, the exit of a VDV coincides with a sudden force release, often followed by a slower buildup. To examine the first feature, Figs. 2(c)–2(f) display the joint distributions of the forces measured by both sensors for the four crowd sizes studied. A strong, symmetric, and positive correlation between the two signals is evident in all cases. After this verification, we

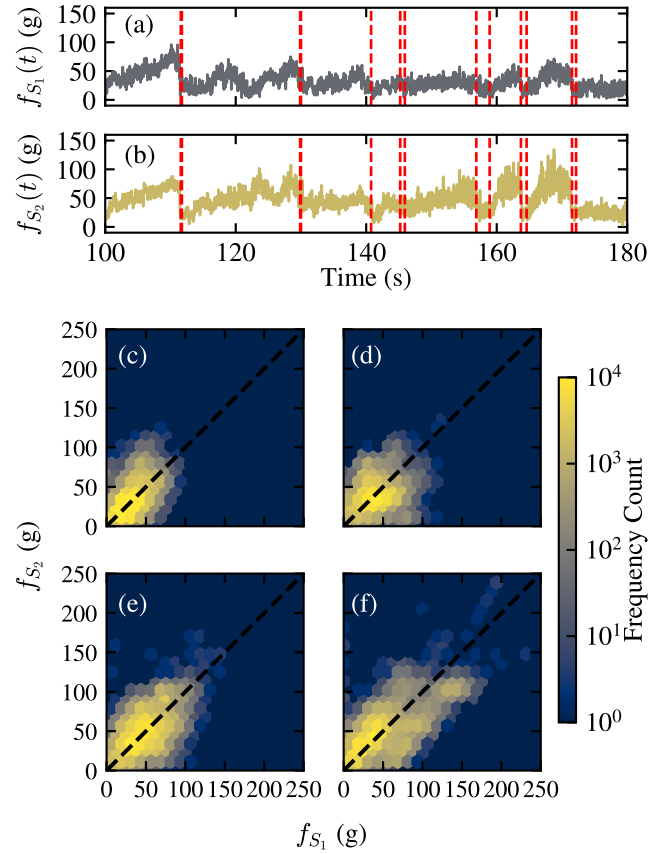


FIG. 2. (a) and (b) Temporal evolution of the force signals measured by sensors S_1 and S_2 for a time window extracted from the experiment with $N_{\text{tot}} = 30$. The vertical lines indicate the exit times of the VDV. (c)–(f) Joint distributions between f_{S_1} and f_{S_2} for $N_{\text{tot}} = 30, 40, 45$, and 50 , respectively. The diagonal line illustrates the identity relation.

decided to combine the datasets from both sensors for the subsequent analysis.

We next analyze the distributions of all recorded forces for the different crowd sizes investigated. As before, to better visualize the tails of the distributions, we compute the complementary cumulative distribution functions (CCDFs) of the forces [Fig. 3(a)]. These curves exhibit a decay that is exponential or steeper, a characteristic feature of granular materials and other discrete systems [22,33]. Additionally, configurations with a larger number of particles, N_{tot} , yield higher overall force values.

To investigate the origin of this dependence, we discriminate the forces developed when the system is bursting (f_b when $\Delta t < \tau_{\text{th}}$) from those developed during arrest time intervals (f_c when $\Delta t > \tau_{\text{th}}$). Figures 3(b) and 3(c) display the corresponding distributions revealing that forces developed during bursts remain similar independently on the number of agents in the arena. On the contrary, forces built up during clogs depend on the crowd size suggesting that the observed differences in the global distributions of Fig. 3(a) primarily originate from the behavior occurring at the flow arrests. Moreover, comparing the distributions obtained in bursting time intervals and arrests (for the same number of agents), we found a feature consistent with the classification of clogged

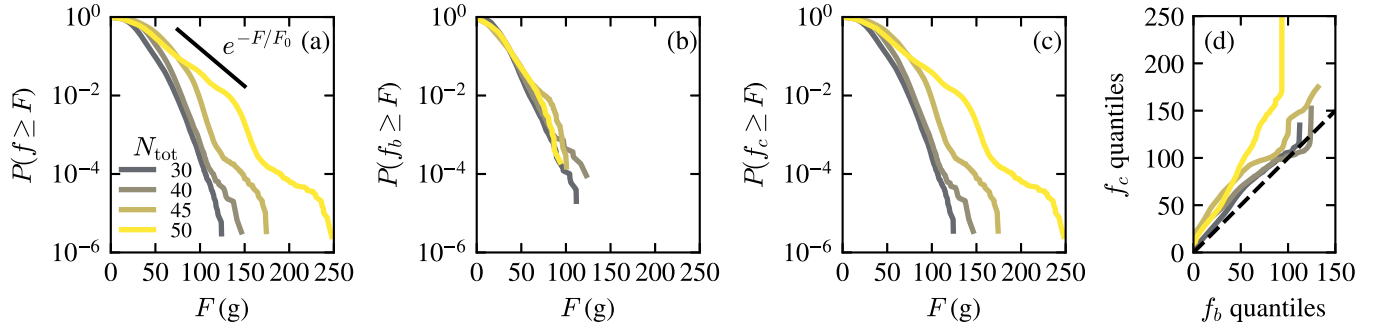


FIG. 3. (a) Complementary CDF of all forces recorded with the two sensors for experiments with different numbers of N_{tot} as indicated in the legend. The black solid line illustrates an exponential trend. (b) and (c) show the complementary CDFs of the forces measured during bursts and clogs, respectively. (d) Quantile-Quantile plot comparing the force distributions during bursts (f_b) and clogs (f_c). The color code in all panels is the same as in (a).

and unclogged states based on the exponent γ shown in Fig. 1(c). For $N_{\text{tot}} = 30$ and 40, where the system is unclogged ($\gamma > 2$), the force distributions remain similar regardless of whether or not there is a clog at the time at which the force is measured. In contrast, for $N_{\text{tot}} = 45$ and 50, where $\gamma \leq 2$, the system enters a clogged regime characterized by a clear difference in the force measurement depending on whether it is taken during the burst or the arrest. This behavior is also evident in the quantile-quantile plot of f_c and f_b [Fig. 3(d)], where the curves for $N_{\text{tot}} = 30$ and 40 lie close to the identity line, indicating similar force distributions. In contrast, the curves for $N_{\text{tot}} = 45$ and 50 show clear deviations, reflecting the development of larger forces during arrests than during bursts.

In principle, this behavior seems to be related with the dynamics already shown in Figs. 2(a) and 2(b), where sudden drops of pressure occur when an agent exits the arena, being often followed by gradual increases in pressure while the system remains clogged. Then, to gain further insight into these dynamics, we have computed the force increments δf over a given time δt , i.e., $\delta f = f(t + \delta t) - f(t)$. As an example, in Figs. 4(a)–4(d) we represent the distributions of δf using $\delta t = 2.5$ s for the four crowd sizes investigated here. The unclogged states ($N_{\text{tot}} = 30$ and 40) yield distributions that are symmetric and narrow (with a width increasing with N_{tot}). By contrast, clogged states ($N_{\text{tot}} = 45$ and 50) exhibit moderate to strong asymmetries, with a predominance of negative δf values. This reflects that force drops are both more frequent and more intense when the system is in a clogged state. To corroborate this result and generalize it, Fig. 4(e) shows the skewness of the δf distributions as a function of δt . The unclogged scenarios ($N_{\text{tot}} = 30$ and 40) maintain skewness values slightly below zero, indicating the existence of nearly symmetric force increment distributions regardless of the time window used to compute them. This can be explained by a negligible pressure accumulation during clogs. On the contrary, for the clogged state ($N_{\text{tot}} = 45$ and 50), the skewness is always negative, confirming the strong asymmetry in the temporal evolution of pressure which is characterized by intense force drops and slow force growth. Interestingly, in the clogged state, the skewness becomes increasingly negative with increasing δt , saturating for $\delta t \approx 3$ s. This value suggests a characteristic timescale during which force release events take place, dom-

inating the dynamics over another, larger timescale, at which the stress accumulation occurs.

To further investigate the temporal evolution of pressure during the flow, we focused on the abrupt force drops observed in Figs. 2(a) and 2(b), which typically coincide with the exit of a VDV. To characterize these events, we segmented the force signal around each exit time, taking t_0 as the instant when a particle leaves the arena. Figure 5(a) shows the average force, $\hat{f}(t)$, obtained by averaging all events aligned at t_0 . In all cases, a sharp force drop occurs at the exit time, followed by a gradual recovery. The amplitude of this drop depends on the total number of VDV: as N_{tot} increases, the magnitude of

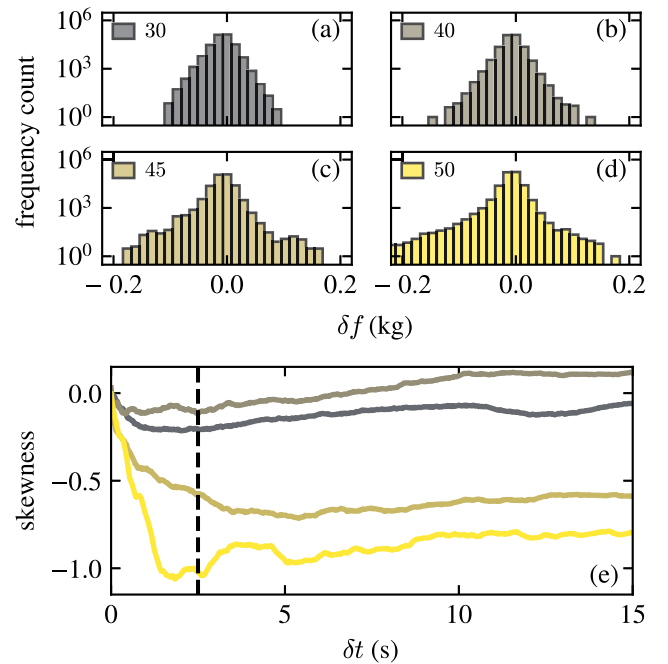


FIG. 4. (a)–(d) Histograms of the distributions of the force change δf in a time interval $\delta t = 2.5$ s for each of the experimental configurations. (e) Skewness of the δf distribution as a function of δt . The vertical line indicates $\delta t = 2.5$ s, the value used in the analysis shown in (a)–(d). The color code in all panels is the same as in (a)–(d).

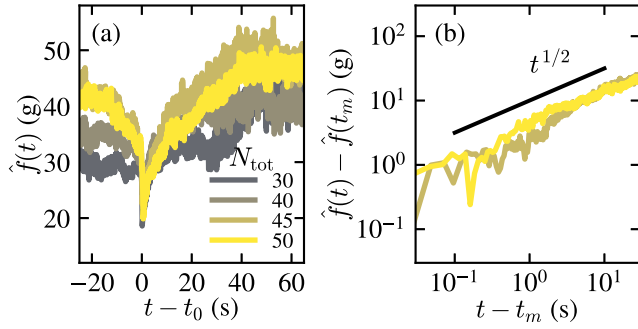


FIG. 5. (a) Average force around exit events for all experimental conditions, where t_0 denotes the precise exit time of a VDV. Each curve represents the mean force $\hat{f}(t)$ obtained by aligning all events at t_0 . (b) Temporal evolution of the force relative to its minimum value $\hat{f}(t_m)$, averaged over all exit events in the clogged regime ($N_{\text{tot}} = 45$ and 50). The color code is the same as in (a).

the drop becomes larger, reflecting stronger stress release and higher pressure accumulation in denser configurations.

To quantify this pressure buildup, we calculated the average force relative to its minimum value $\hat{f}(t_m)$ around each exit and shifted the time axis by setting t_m as the reference. This alignment allows all events to be compared at their point of minimum force, emphasizing the subsequent evolution of the buildup process. The results, shown in Fig. 5(b) for the denser cases ($N_{\text{tot}} = 45$ and 50), reveal that the mean pressure growth follows a power-law dependence with an exponent close to $1/2$. This indicates that the pressure increases at a decelerating rate as time progresses after the release event. Such scaling is consistent with diffusionlike relaxation or structural aging processes, in which the system progressively reorganizes under internal stress. These observations suggest that the collective rearrangements governing clog formation and dissolution are ruled by slow, correlated dynamics—compatible with the aging mechanisms previously proposed to explain the broad-tailed distributions of clogging durations [14,15].

IV. CONCLUSIONS

In this work, we present an experimental study of the mechanical pressure dynamics in the flow of self-propelled

particles through a narrow opening. Our results show that pressure dynamics is strongly correlated with the system's dynamical phase, i.e., clogged or unclogged. In the clogged regime, there is a net buildup of pressure after the passage of each agent, leading to a clear difference among pressures during bursts and arrests. Furthermore, the distribution of pressure increments becomes asymmetric, with a marked predominance of large negative fluctuations. In contrast, in the unclogged regime, the pressure buildup becomes less evident, implying no difference among pressures during bursting and arrest intervals, and symmetric distributions of pressure increments.

Strikingly, when the system is clogged, the pressure grows with time following a square-root dependence. This scaling is consistent with a structural aging process, previously proposed as a mechanism underlying the broad, power-law-like distribution of clogging intervals. These findings suggest that the temporal evolution of mechanical pressure can serve as a reliable indicator of the system's state, offering an alternative to more tedious methods based on bursts and arrest intervals measurements.

ACKNOWLEDGMENTS

This work was funded by Project PICTO-2022-ITBA-00001 (Agencia Nacional de Promoción Científica y Tecnológica, Argentina and Instituto Tecnológico de Buenos Aires, Argentina) and Grant PID2023-146422NB-I00 funded by MICIU/AEI/10.13039/501100011033 and by ERDF/UE (Ministerio de Ciencia e Innovación, Agencia Estatal de Investigación and European Union).

DATA AVAILABILITY

The data that support the findings of this article are openly available [30].

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